

Vulnerability assessment

OpenVAS/Greenbone

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Contents

1	OpenVAS Scan	2
1.1	OpenVAS Scan	2
1.1.1	Configuration → Ports	3
1.1.2	Target configuration	6
1.1.3	Starting the scan	11
2	AI Usage Declaration	15

Chapter 1

OpenVAS Scan

In this chapter the most important aspects of Greenbone OpenVAS are explained, with practical examples to give the reader an understanding of the tools usage.

1.1 OpenVAS Scan

In this section it will be explained how to start a scan (that could be just for a host, or a group of hosts). We will create a target, which means declaring which host (or group of hosts) to be scanned, which port range and protocol (TCP or UDP).

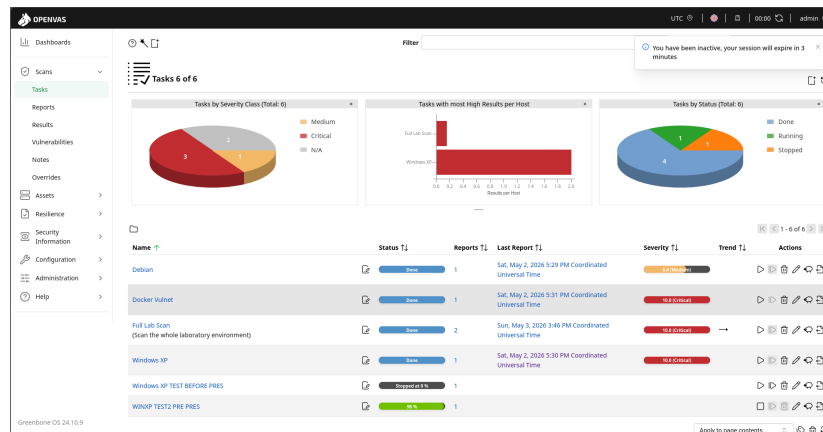
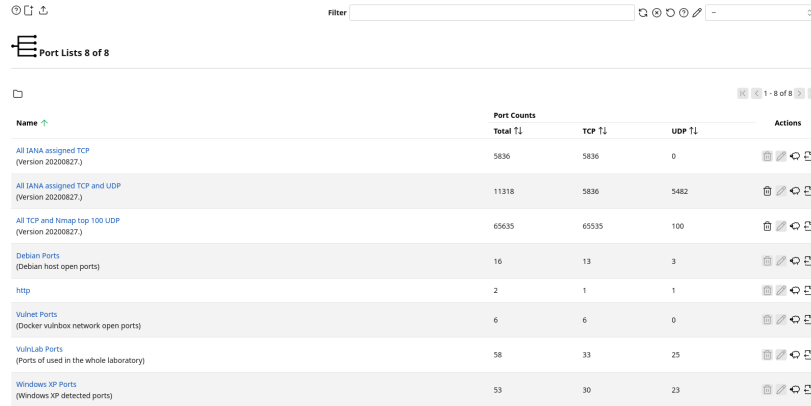


Figure 1.1: Scan windows

This is the scan window, where you will create the tasks which are configurations for the targets and ports that are going to be scanned, how many NVTs (Network Vulnerability Test) are going to be used for the scan (done in

the Scan config windows under Configuration), how many concurrent hosts are being scanned at a given moment etc. We will get to all of the configuration in time.

1.1.1 Configuration → Ports



Name	Port Counts			Actions
	Total T/U	TCP T/U	UDP T/U	
All IANA assigned TCP (Version 20200627)	5836	5836	0	[Add] [Edit] [Delete]
All IANA assigned TCP and UDP (Version 20200627)	11318	5836	5482	[Add] [Edit] [Delete]
All TCP and Nmap top 100 UDP (Version 20200627)	65635	65535	100	[Add] [Edit] [Delete]
Debian Ports (Debian host open ports)	16	13	3	[Add] [Edit] [Delete]
http	2	1	1	[Add] [Edit] [Delete]
Vulnet Ports (Docker vulnbox network open ports)	6	6	0	[Add] [Edit] [Delete]
VulnLab Ports (Ports of used in the whole laboratory)	58	33	25	[Add] [Edit] [Delete]
Windows XP Ports (Windows XP detected ports)	53	30	23	[Add] [Edit] [Delete]

Figure 1.2: Port creation interface

This is the ports creation interface, you can create custom ports ranges looking for services that use TCP/UDP. For the most part OpenVAS works by giving us templates, in this case, they are for a complete scan of all IANA (Internet Assigned Numbers Authority) ports, which means it will scan for all 65536 ports only TCP, UDP or both. It really just depends on which preset is chosen.

OpenVAS encodes ports in a format like **T:1 – 1023, 4500,U:2 – 100, 3246** which means that it will scan for TCP ports between 1 and 1023 and 4500 alone, and for UDP ports from 2 to 100 and 3246 alone. In this way you can define ports range and choose to do only TCP or UDP scans indicated by **T:** or **U:**.

Under the hood OpenVAS uses Nmap to scan the ports, and just like it UDP ports scan will take considerably longer than TCP. This is because when a TCP request is made, the server either accept the connection or it rejects it, while for UDP being a connectionless protocol will wait until timeout. So beware about this aspect.

For our experiments we will create two presets: one for ftp service (port 21) and one for an http server (port 80). Go on the add port icon on the top of the interface (the little square with a plus), and you should get this popup.

New Port List

×

Name

WIN XP ONLY FTP

Comment

Port Ranges

☒ Manual

T:21

☐ From file

Cancel

Save

New Port List

×

Name

WIN XP FTP&HTTP

Comment

Port Ranges

☒ Manual

T:21,80

☐ From file

Cancel

Save

(a) Port configuration limited to TCP 21 (FTP). (b) Port configuration limited to TCP 21 and 80 (FTP).

Figure 1.3: Ports configuration

Those configurations are going to be used later to scan a Windows XP target with FTP and HTTP servers.

Credentials for target access

The screenshot shows the OpenVAS web interface with the 'Credentials' section selected in the left sidebar. The main area displays a table of 10 credentials. A notification at the top right states: 'You have been inactive, your session will expire in 3 minutes'.

Name	Type	Allow insecure use	Login	Actions
Admin	Username + Password	Yes	admin	[Edit] [Delete] [Add]
Admin vulnbox	Username + Password	Yes	admin	[Edit] [Delete] [Add]
Debian	Username + Password	Yes	debian	[Edit] [Delete] [Add]
Guest	Username + Password	Yes	guest	[Edit] [Delete] [Add]
Manager	Username + Password	Yes	manager	[Edit] [Delete] [Add]
MySQL vulnbox (vulnbox:mysql password)	Username + Password	Yes	vulnbox	[Edit] [Delete] [Add]
Postgres	Username + Password	Yes	postgres	[Edit] [Delete] [Add]
Root	Username + Password	Yes	root	[Edit] [Delete] [Add]
Tomcat	Username + Password	Yes	tomcat	[Edit] [Delete] [Add]
User	Username + Password	Yes	user	[Edit] [Delete] [Add]

At the bottom right, there is a button 'Apply to page contents' and a pagination indicator '1 - 10 of 10'.

Figure 1.4: Credentials window

Credentials are used for deeper scans, you give the credentials to OpenVAS and it will try to login to the machine, scanning the host from within, looking at configuration files, installed packages. This is done to improve scan performance since it flags problems that are not visible from the outside, but may lead to exploitable vulnerabilities. This function can also be used to check for default credentials if they are in use or no.

New Credential

×

Name

Test

Comment

Type

Username + Password

Allow insecure use

☐ Yes
☒ No

Auto-generate

☐ Yes
☒ No

Username

Password

Cancel

Save

New Credential

×

Name

Test

Comment

Type

Username + Password

Username + SSH Key

SNMP

SMIME Certificate

PGP Encryption Key

Password only

Client Certificate

Cancel

Save

(a) Credentials Configuration Step 1

(b) Credentials Configuration Step 2

Figure 1.5: Configuration for the FTP only target

As you can see many possibilities are available, setting both username and password, ssh access, certificates etc. But for our example this feature will not be used. This feature if used not only will report problems, but it will also write in the report if it had succeeded or failed to log onto the host.

Host	Protocol	Result	Port / User
192.168.56.102 - WINDOWS-XP	SSH	Failure	Protocol SSH, Port 22, User debian : Login failure
192.168.56.102 - WINDOWS-XP	SMB	Failure	Protocol SMB, Port 445, User debian
172.30.2.2	SSH	Failure	Protocol SSH, Port 22, User debian : Login failure

5

Host	Protocol	Result	Port/User
192.168.56.104	SSH	Failure	Protocol SSH, Port 22, User debian : Login failure
172.30.3.2	SSH	Failure	Protocol SSH, Port 22, User debian : Login failure
172.30.2.3	SSH	Failure	Protocol SSH, Port 22, User debian : Login failure
172.30.2.1	SSH	Failure	Protocol SSH, Port 22, User debian : Login failure
172.30.1.2	SSH	Failure	Protocol SSH, Port 22, User debian : Login failure
172.30.0.1 - DEBIAN	SSH	Failure	Protocol SSH, Port 22, 'su' User root : Switch user failed
172.30.0.1 - DEBIAN	SMB	Success	Protocol SMB, Port 445, User debian
192.168.56.101 - DEBIAN	SSH	Failure	Protocol SSH, Port 22, 'su' User root : Switch user failed
192.168.56.101 - DEBIAN	SMB	Success	Protocol SMB, Port 445, User debian
172.30.3.1	SSH	Failure	Protocol SSH, Port 22, User debian : Login failure
172.30.3.1	SMB	Success	Protocol SMB, Port 445, User debian
172.30.2.4	SSH	Failure	Protocol SSH, Port 22, User debian : Login failure
172.30.2.4	SMB	Success	Protocol SMB, Port 445, User debian
172.30.1.1	SSH	Failure	Protocol SSH, Port 22, User debian : Login failure
172.30.0.2	SSH	Failure	Protocol SSH, Port 22, User debian : Login failure

Figure 1.6: Example of login attempts in the report exported as PDF in the Full Lab Scan provided in the OpenVAS VM

1.1.2 Target configuration

Name	Hosts	IPs	Port List	Credentials	Actions
Debian Target (Debian VM vulnbox)	192.168.56.103	1	Debian Ports	SSH:Debian SSH:Elevate:Root SMB:NTLM:Debian	[Icons]
VM Targets (VMs in the laboratory environment)	192.168.56.101-192.168.56.104	4	VulnLab Ports	SSH:Debian SSH:Elevate:Root SMB:NTLM:Debian	[Icons]
vulnbox1 (LongShelf)	172.30.0.2	1	VulnLab Ports	SSH:Admin SSH:Elevate:Root	[Icons]
vulnbox2 (Apache Struts)	172.30.1.2	1	VulnLab Ports	SSH:Tomcat SSH:Elevate:Root	[Icons]
vulnbox3 (OWASP)	172.30.2.2	1	VulnLab Ports	SSH:JSP/SQL_vulnbox3 SSH:Elevate:Root	[Icons]
vulnbox4 (SSRF)	172.30.2.3	1	VulnLab Ports	SSH:User SSH:Elevate:Admin	[Icons]
vulnbox5 (Samba)	172.30.2.4	1	VulnLab Ports	SSH:Guest SSH:Elevate:Root SMB:NTLM:Guest	[Icons]

Figure 1.7: Target configuration interface

This is the target creation interface, which is used to create all targets that are going to be scanned expressed in single IP or a range of IPs (e.g. 192.168.56.101-192.168.56.105).

Before scanning the Windows XP server you would want to go to the terminal of the machine and type **ipconfig /all**, copy the IP (which should be along the lines of 192.168.56.10X).

```

C:\Documents and Settings\windows>ipconfig /all

Windows IP Configuration

Host Name . . . . . : windows-xp
Primary Dns Suffix . . . . . :
Node Type . . . . . : Unknown
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) PRO/1000 T Server Adapter
Physical Address. . . . . : 08-00-27-BB-85-A1
Dhcp Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IP Address. . . . . : 192.168.56.104
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
DHCP Server . . . . . : 192.168.56.100
Lease Obtained. . . . . : Tuesday, May 12, 2026 2:35:15 PM
Lease Expires . . . . . : Tuesday, May 12, 2026 2:45:15 PM

C:\Documents and Settings\windows>

```

Figure 1.8: Terminal output of ipconfig command

After the IP was discovered, its time to configure the target like so:

(a) FTP Target Step 1

(b) FTP Target Step 2

Figure 1.9: Configuration for the FTP only target

In this popup you will set target name, the IP (or range of IPs), the excluded hosts in case you have a range of IPs that are not to be scanned, the port list preset made before and the credentials presets that are set in the credentials window (not used in this example).

Now configure the targets also for the FTP and HTTP target like so:

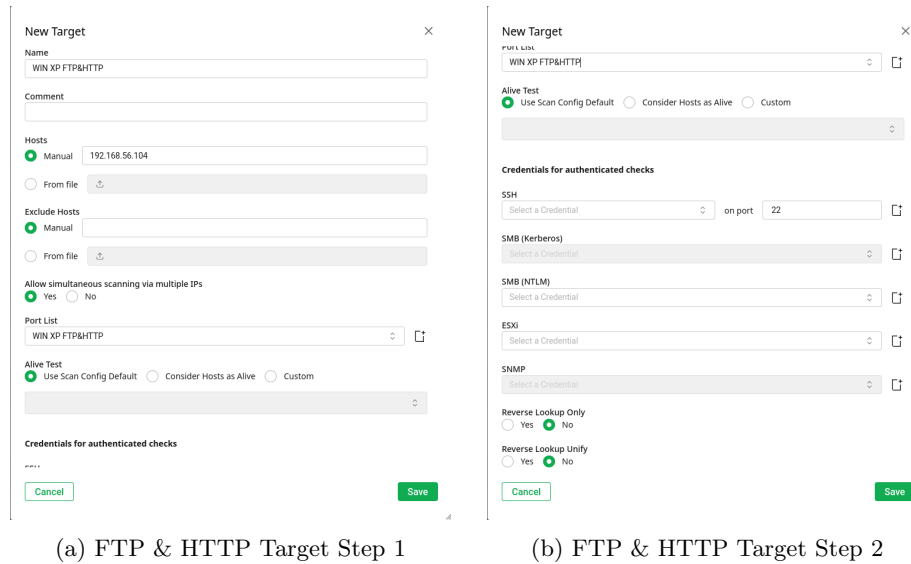


Figure 1.10: Configuration for the FTP & HTTP target

Remember the difference between these two scans is that for the first one we are searching only for the FTP service, while for the second we are searching for FTP and HTTP web server.

Scan configs

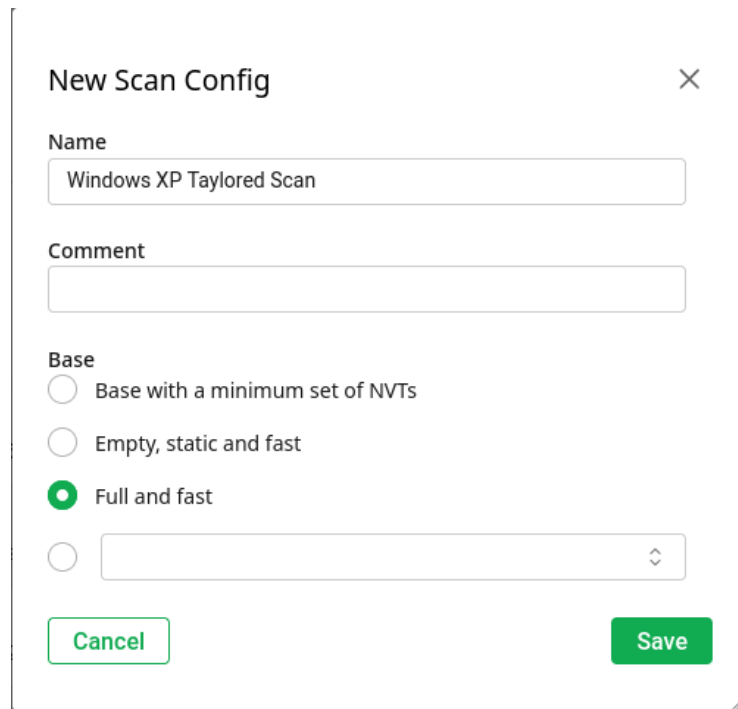
Figure 1.11 is a screenshot of the OpenVAS 'Scan Configs' window. The window shows a list of scan configurations with columns for Name, Family, Total T1, Trend T1, NVTs, and Actions. The configurations are listed in a table.

Name	Family	Total T1	Trend T1	NVTs	Actions
Base (Base configuration template with a minimum set of NVTs required for a scan. Version 20200827)	2	→	3	→	[Icons]
Discovery (Network Discovery scan configuration. Version 20201215)	11	→	3320	→	[Icons]
empty (Empty and static configuration template. Version 20201215)	0	→	0	→	[Icons]
Full and fast (Most NVTs optimized by using previously collected information. Version 20201215)	60	→	176190	→	[Icons]
Full and fast Clone 1 (Most NVTs optimized by using previously collected information. Version 20201215)	42	→	15600	→	[Icons]
Full and fast Clone 1 Clone 1 (Most NVTs optimized by using previously collected information. Version 20201215)	42	→	15600	→	[Icons]
Host Discovery (Network Host Discovery scan configuration. Version 20201215)	2	→	2	→	[Icons]
Host Discovery Clone 1 (Network Host Discovery scan configuration. Version 20201215)	2	→	2	→	[Icons]
Log4jShell (Configuration with checks for Log4j and CVE-2021-44228. Version 20211227)	10	→	29	→	[Icons]

Figure 1.11: Scan configs window

The scan configuration tells OpenVAS that to scan for and how to scan it. Essentially the Full and Fast preset looks at everything possible from Cisco appliances, Fedora security, Windows services etc. This is the reason why the database is so big, occupying tens of gigabytes. Since there are thousands of

settings, to save time we will configure a pre-set at a high level tailoring it for Windows XP.

A screenshot of a 'New Scan Config' dialog box. The dialog has a title bar with a close button (X) in the top right corner. Inside, there are three main sections: 'Name' with a text input field containing 'Windows XP Taylored Scan'; 'Comment' with an empty text input field; and 'Base' with three radio button options: 'Base with a minimum set of NVTs', 'Empty, static and fast', and 'Full and fast'. The 'Full and fast' option is selected, indicated by a green dot. Below these options is a dropdown menu with a small upward arrow icon. At the bottom of the dialog are two buttons: 'Cancel' on the left and 'Save' on the right. The 'Save' button is green with white text, while 'Cancel' is a light green outline.

New Scan Config

Name

Windows XP Taylored Scan

Comment

Base

☐ Base with a minimum set of NVTs

☐ Empty, static and fast

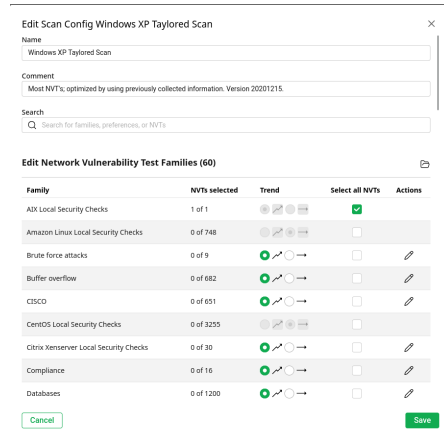
☒ Full and fast

☐ [Dropdown]

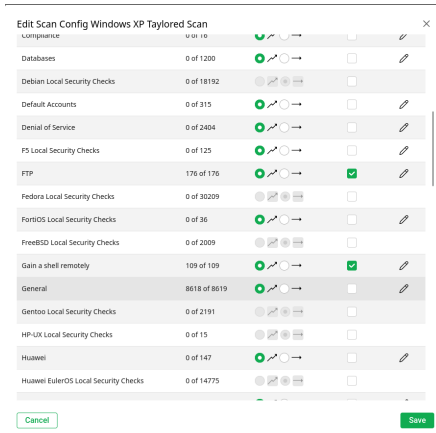
Cancel Save

Figure 1.12: Tailored scan creation

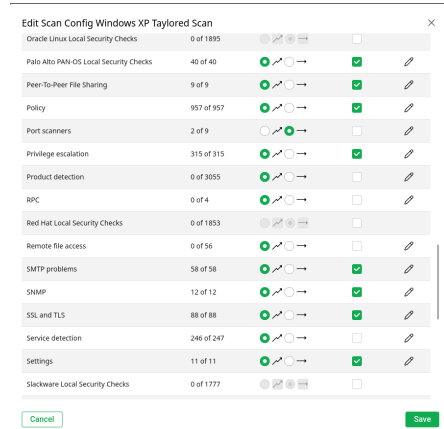
To create it select the Full and Fast preset, save it and then edit it by clicking to the pencil icon, then deselect everything that is not in the following.



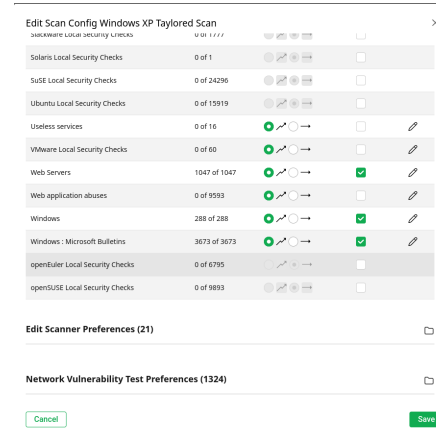
(a) FTP & HTTP Tailored scan options
1



(b) FTP & HTTP Tailored scan options
2



(a) FTP & HTTP Tailored scan options
3



(b) FTP & HTTP Tailored scan options
4

Figure 1.14: Configuration for the FTP & HTTP Tailored scan

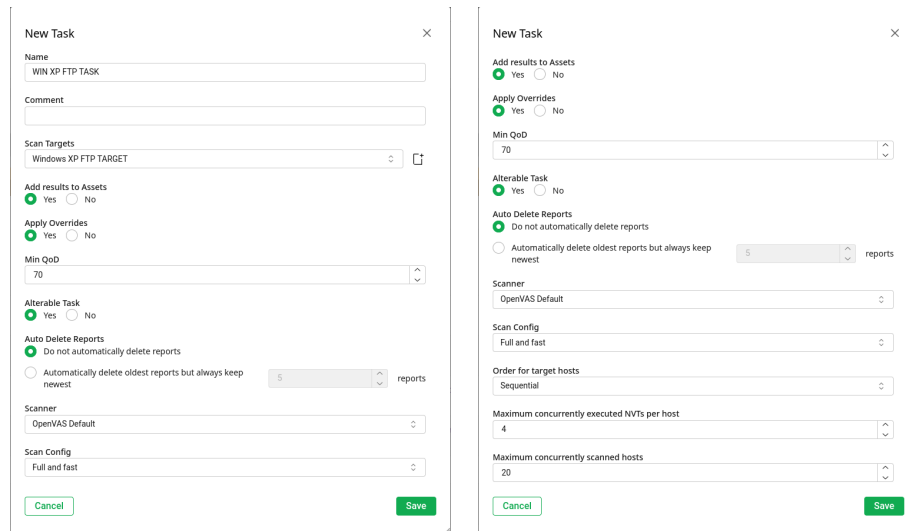
This is far from being a completely optimized scan for our purpose, since we are able to better tailor each section seen here by pressing on the pencil mark, but that would become time consuming since you would select/deselect hundreds of different things it scans, so for simplicity it is ignored in this experiment.

We selected the Full and Fast preset because for each category of scan there are hundreds or even thousands of possible subconfiguration. If you are curious about it just click the edit icon in the scan configuration popup that will show you how many configurations you can activate/deactivate. These suboptions

are omitted from the document since it would be too time consuming to explore them, but are left as an exercise to the reader to further explore.

1.1.3 Starting the scan

Now we are ready to begin the scan, to do this go back to the tasks windows, click on the task add button and add all the settings created so far like this:



The image shows two side-by-side screenshots of a 'New Task' dialog box. Both screenshots have a close button (X) in the top right corner. Screenshot (a) is labeled '(a) FTP task 1' and shows the following settings: Name: WIN XP FTP TASK, Comment: (empty), Scan Targets: Windows XP FTP TARGET, Add results to Assets: Yes (checked), Apply Overrides: Yes (checked), Min QoD: 70, Alterable Task: Yes (checked), Auto Delete Reports: Do not automatically delete reports (checked), Scanner: OpenVAS Default, Scan Config: Full and fast. Screenshot (b) is labeled '(b) FTP task 2' and shows the following settings: Add results to Assets: Yes (checked), Apply Overrides: Yes (checked), Min QoD: 70, Alterable Task: Yes (checked), Auto Delete Reports: Do not automatically delete reports (checked), Scanner: OpenVAS Default, Scan Config: Full and fast, Order for target hosts: Sequential, Maximum concurrently executed NVTs per host: 4, Maximum concurrently scanned hosts: 20. Both screenshots have 'Cancel' and 'Save' buttons at the bottom.

(a) FTP task 1

(b) FTP task 2

This task will scan Windows XP for the FTP service. The options are pretty much self explanatory since you are going to use all the presets set before, the important flag to enable is the alterable task flag just for our convenience. This flag permits you to modify the task at will, otherwise if set to **no** it would not permit you to alter the task once created, and thus you will have to redo everything. This flag is normally set to false because when you scan a system for vulnerabilities or problems, you would want to correct them, then to verify if the problems have been mitigated you would simply clone the task that you previously executed, start it without modifications (if the alterable flag was true then you would be able to modify it) and then check if all problems are fixed. If you still have problems to be fixed then you should repeat until there are no more.

Now you should have a new task created with the name you gave it, in this case **WIN XP FTP TASK** like so:



Now press the play button to queue it so that it will start, it will take approximately 10 minutes to complete. After the scan is done it should appear

in the report tab. Reports are the result of the scan activity, there you will find all the information regarding the scan done, such as host, open ports, vulnerabilities found with its severity score:

Information	Results (1 of 13)	Hosts (1 of 1)	Ports (1 of 1)	Applications (1 of 1)	Operating Systems	CVEs (2 of 0)	Closed CVEs (0 of 0)	TLS Certificates (0 of 0)	Error Messages (0 of 0)	User Tags (0)
1 - 3 of 3										
Vulnerability T1	Severity	Host IP T1	Name T1	Location T1	EPSS Score T1	Percentile T1	Created T1			
Anonymous FTP Login Reporting	1.00 Critical 80 %	192.168.56.104		21/tcp	N/A	N/A	Tue, May 12, 2026 8:08 PM Coordinated Universal Time			
FTP Unencrypted Cleartext Login	0.85 High 70 %	192.168.56.104		21/tcp	N/A	N/A	Tue, May 12, 2026 8:08 PM Coordinated Universal Time			
ICMP Timestamp Reply Information Disclosure	0.80 High 80 %	192.168.56.104		generalsmp	N/A	N/A	Tue, May 12, 2026 8:08 PM Coordinated Universal Time			

This is what we should see in the report tab, on the scan. These are the problems found by OpenVAS, on the FTP service, mainly it tells us that FTP is unsecure and that it has the anonymous user enabled. In the ports window we can see that it successfully detected port 21.

[illegible]

Now we will execute the tailored scan for both FTP and HTTP. The configuration is similar to the previous one, just change the target to the right one and select the tailored scan, remember the alterable task flag.

New Task

Name

WIN XP HTTP&FTP TASK

Comment

Scan Targets

WIN XP FTP&HTTP

Add results to Assets

☒ Yes
☐ No

Apply Overrides

☒ Yes
☐ No

Min QoD

70

Alterable Task

☒ Yes
☐ No

Auto Delete Reports

☒ Do not automatically delete reports
☐ Automatically delete oldest reports but always keep newest 5 reports

Scanner

OpenVAS Default

Scan Config

Windows XP Tailored Scan

Cancel

Save

(a) FTP & HTTP Tailored task 1

New Task

Add results to Assets

☒ Yes
☐ No

Apply Overrides

☒ Yes
☐ No

Min QoD

70

Alterable Task

☒ Yes
☐ No

Auto Delete Reports

☒ Do not automatically delete reports
☐ Automatically delete oldest reports but always keep newest 5 reports

Scanner

OpenVAS Default

Scan Config

Windows XP Tailored Scan

Order for target hosts

Sequential

Maximum concurrently executed NVTs per host

4

Maximum concurrently scanned hosts

20

Cancel

Save

(b) FTP & HTTP Tailored task 2

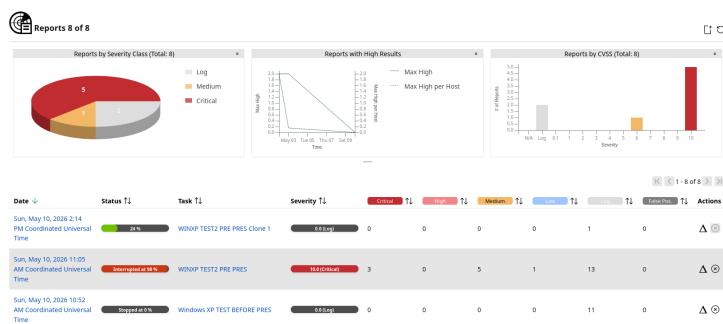
Once the scan is done, you should see something like this if you have configured correctly the Scan configuration and ports:

Vulnerability T1	Severity	QoD T1	Host IP T1	Name T1	Location T1	EPSS Score T1	Percentile T1	Created T1
Microsoft IIS 3.0k SOAP Filter Applied - Active Check	High	70 %	192.168.56.104		80tcp	N/A	N/A	Tue, May 12, 2020 8:32 PM Coordinated Universal Time
Microsoft Internet Information Services (IIS) End of Life (EoL) Detection	High	80 %	192.168.56.104		80tcp	N/A	N/A	Tue, May 12, 2020 8:32 PM Coordinated Universal Time
Anonymous FTP Login Reporting	Medium	80 %	192.168.56.104		21tcp	N/A	N/A	Tue, May 12, 2020 8:25 PM Coordinated Universal Time
HTTP Debugging Methods (TRACE/TRACE) Enabled	High	99 %	192.168.56.104		80tcp	N/A	N/A	Tue, May 12, 2020 8:28 PM Coordinated Universal Time
Microsoft IIS Default Welcome Page Information Disclosure Vulnerability	High	70 %	192.168.56.104		80tcp	N/A	N/A	Tue, May 12, 2020 8:32 PM Coordinated Universal Time

The scan not only shows the FTP problems you have found before, but it also alerts you about the Internet Information Services (IIS) End of Life (EoL); its vulnerabilities might be used by an attacker, and that the Indexing Service filter of the web server is enabled, which can expose sensitive information about the server and its content, and it might give to an attacker remote system level access to the machine. These are just some of the vulnerabilities of the scans that were found using the tailored scan.

Those are just a few of the vulnerabilities found, many more are to be found doing a full scan of the host across all ports using the Full and Fast Scan config preset, which is left as an exercise to the reader.

Report



This is the report tab, at the end of the scan you can visualize the result of the scans, export it in pdf by clicking on the interested report, then the download icon on top of the page and it will open a popup to download it in various formats, with PDF being the best option for visualization. The report does not serve only as a list of the vulnerabilities found, but also it gives the user the mitigation step to fix a certain vulnerability, or at least point in the right direction.

Chapter 2

AI Usage Declaration

This report was prepared with the assistance of AI tools for drafting and editing support. The structure and the core contents of the report were designed by the author. All final content was reviewed and approved by the author.